Inorganic Cumulative Examination

February 17, 2012

The ability to synthesize small molecules can be important in mechanistic studies, isotopic labeling and the synthesis of larger molecules. In addition, it provides a fundamental look at some of the chemistry. In that context, provide a small-scale laboratory synthesis using common laboratory reagents for the following small molecules. Getting the material from a lecture cylinder is not considered synthesis.

1. H_2
2. O_2
3. N_2
4. Cl_2
5. NH_3
6. CO
7. N_2O
8. NO
9. CO_2
10. HCl
11. NO_2
12. SO_2
13. PH_3
14. H_2S
15. B_2H_6